四　向量组的等价

定 义　设$\alpha_1,\alpha_2,\cdots,\alpha_s$与$\beta_1,\beta_2,\cdots,\beta_t$是$F^n$中的两个向量组．如果$\alpha_1,\alpha_2,\cdots,\alpha_s$可由$\beta_1,\beta_2,\cdots,\beta_t$线性表示，$\beta_1,\beta_2,\cdots,\beta_t$也可以由$\alpha_1,\alpha_2,\cdots,\alpha_s$线性表示，那么称$\alpha_1,\alpha_2,\cdots,\alpha_s$与$\beta_1,\beta_2,\cdots,\beta_t$**等价**，记作

$$\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\cong\{\beta_1,\beta_2,\cdots,\beta_t\}.$$

命 题 7 向量组的等价满足

(1) **对称性**：如果 $\{\alpha_1, \alpha_2, \cdots, \alpha_s\} \cong \{\beta_1, \beta_2, \cdots, \beta_t\}$，那么

$$\{\beta_1, \beta_2, \cdots, \beta_t\} \cong \{\alpha_1, \alpha_2, \cdots, \alpha_s\};$$

(2) **传递性**：如果 $\{\alpha_1, \alpha_2, \cdots, \alpha_s\} \cong \{\beta_1, \beta_2, \cdots, \beta_t\}$，并且 $\{\beta_1, \beta_2, \cdots, \beta_t\} \cong \{\gamma_1, \gamma_2, \cdots, \gamma_u\}$，那么

$$\{\alpha_1, \alpha_2, \cdots, \alpha_s\} \cong \{\gamma_1, \gamma_2, \cdots, \gamma_u\}.$$

例 13 确定下列向量组是否等价：

$$(1)\ \varepsilon_1=\begin{pmatrix}1\\0\end{pmatrix},\ \varepsilon_2=\begin{pmatrix}0\\1\end{pmatrix} \text{与}\ \alpha_1=\begin{pmatrix}1\\1\end{pmatrix},\ \alpha_2=\begin{pmatrix}2\\-1\end{pmatrix};$$ 等价

$$(2)\ \varepsilon_1=\begin{pmatrix}1\\0\end{pmatrix},\ \varepsilon_2=\begin{pmatrix}0\\1\end{pmatrix} \text{与}\ \beta_1=\begin{pmatrix}2\\0\end{pmatrix},\ \beta_2=\begin{pmatrix}-1\\0\end{pmatrix}.$$ 不等价

引理1 设$\alpha_1,\alpha_2,\cdots,\alpha_s$与$\beta_1,\beta_2,\cdots,\beta_t$是$F^n$中的两个向量组. 如果$\alpha_1,\alpha_2,\cdots,\alpha_s$可由$\beta_1,\beta_2,\cdots,\beta_t$线性表示, 那么 $L(\alpha_1,\alpha_2,\cdots,\alpha_s)\subseteq L(\beta_1,\beta_2,\cdots,\beta_t)$.

证明 设$\alpha_1,\alpha_2,\cdots,\alpha_s$可由$\beta_1,\beta_2,\cdots,\beta_t$线性表示. 因为$L(\alpha_1,\alpha_2,\cdots,\alpha_s)$中的任意向量 $\alpha=k_1\alpha_1+k_2\alpha_2+\cdots+k_s\alpha_s$ 都可由$\beta_1,\beta_2,\cdots,\beta_t$线性表示, 所以$\alpha\in L(\beta_1,\beta_2,\cdots,\beta_t)$. 因此, $L(\alpha_1,\alpha_2,\cdots,\alpha_s)\subseteq L(\beta_1,\beta_2,\cdots,\beta_t)$.

命题 8 如果$\alpha_1,\alpha_2,\cdots,\alpha_s$与$\beta_1,\beta_2,\cdots,\beta_t$是$F^n$中的两个向量组, 那么$\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\cong\{\beta_1,\beta_2,\cdots,\beta_t\}$的充分必要条件是 $L(\alpha_1,\alpha_2,\cdots,\alpha_s)=L(\beta_1,\beta_2,\cdots,\beta_t)$.

证明 根据引理1，命题的必要性成立.

充分性. 设$L(\alpha_1,\alpha_2,\cdots,\alpha_s)=L(\beta_1,\beta_2,\cdots,\beta_t)$.

因为$\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\subset L(\beta_1,\beta_2,\cdots,\beta_t)$，所以

$\alpha_1,\alpha_2,\cdots,\alpha_s$可以由$\beta_1,\beta_2,\cdots,\beta_t$线性表示. 对称地,

$\beta_1,\beta_2,\cdots,\beta_t$ 可以由$\alpha_1,\alpha_2,\cdots,\alpha_s$线性表示.

3.3　极大无关组与向量组的秩

一　向量组的秩

定义 设$\alpha_1,\alpha_2,\cdots,\alpha_t$是$F^n$中的一个向量组, $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$是$\alpha_1,\alpha_2,\cdots,\alpha_t$中的部分向量构成的向量组. 如果

(1) $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$ 是线性无关的,

(2) 对$\alpha_1,\alpha_2,\cdots,\alpha_t$中的任意向量$\alpha_k$, 向量组 $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r},\alpha_k$ 都是线性相关的, 那么

称$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$是$\alpha_1,\alpha_2,\cdots,\alpha_t$的**极大线性无关组**,

简称为**极大无关组**.

例 14 设 $\alpha_1=\begin{pmatrix}-1\\0\end{pmatrix},\alpha_2=\begin{pmatrix}3\\1\end{pmatrix},\alpha_3=\begin{pmatrix}1\\1\end{pmatrix},\alpha_4=\begin{pmatrix}2\\-1\end{pmatrix}$.

显然α_1,α_2与α_2,α_3都是$\alpha_1,\alpha_2,\alpha_3,\alpha_4$的极大无关组.

所以，一般来说,向量组的极大无关组是不唯一的.

极大无关组 $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$ $\begin{cases} \alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r} \text{ 线性无关} \\ \alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r},\alpha_k \text{ 线性相关，} \forall k=1,\cdots t \end{cases}$

$\xrightarrow{1}$ α_k 可由 $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$ 线性表示

$\xrightarrow{2}$ $\{\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}\}\cong\{\alpha_1,\alpha_2,\cdots,\alpha_t\}$

$\xrightarrow{3}$ $\left.\begin{array}{l} \alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s} \text{ 也是极大无关组} \\ \{\alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s}\}\cong\{\alpha_1,\alpha_2,\cdots,\alpha_t\} \end{array}\right\}$

$\xrightarrow{4}$ $\{\alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s}\}\cong\{\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}\}$

线性无关 $\xrightarrow{5}$ $s\le r$

线性无关 $\xrightarrow{6}$ $r\le s$

$\xrightarrow{7}$ $r=s$ $\xrightarrow{8}$ 极大无关组中向量个数唯一 $\xrightarrow[\text{定义}]{9}$ **向量组的秩**

$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$ $\xrightarrow{10}$ 可线性表示

$\underbrace{\{\alpha_1,\alpha_2,\cdots,\alpha_t\}\text{中任意 } r+1 \text{ 个向量}}$ $\xrightarrow{11}$ 线性相关

（否则 $r+1\le r$，矛盾）

$\xrightarrow{12}$ $\{\alpha_1,\alpha_2,\cdots,\alpha_t\}$中任意 r **个线性无关的向量**都构成**极大无关组**

$\xrightarrow{13}$

存在 r 个线性无关的向量

向量组中任意 $r+1$ 个向量线性相关

$\xLeftrightarrow{14}$ $\mathrm{r}\{\alpha_1,\alpha_2,\cdots,\alpha_t\}=r$

命题9 如果$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$是$\alpha_1,\alpha_2,\cdots,\alpha_t$的极大无关组，那么$\alpha_1,\alpha_2,\cdots,\alpha_t$中的任意向量$\alpha_k$都可由$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$线性表示.

命题10 向量组与它的极大无关组是等价的.

命题11 向量组$\alpha_1,\alpha_2,\cdots,\alpha_t$的极大无关组中向量的个数是唯一的.

证明 设$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$与$\alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s}$都是$\alpha_1,\alpha_2,\cdots,\alpha_t$的极大无关组，那么根据命题10，$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$与$\alpha_1,\alpha_2,\cdots,\alpha_t$等价，$\alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s}$也与$\alpha_1,\alpha_2,\cdots,\alpha_t$等价.

根据命题7，$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$与$\alpha_{j_1},\alpha_{j_2},\cdots,\alpha_{j_s}$等价.

根据定理3的推论1，得到$r=s$.

定义 向量组$\alpha_1,\alpha_2,\cdots,\alpha_t$的极大无关组中向量的个数称为**向量组的秩**，记作$r\{\alpha_1,\alpha_2,\cdots,\alpha_t\}$.

命题12　向量组的秩为r的充分必要条件是

(1) 向量组中存在r个线性无关的向量;

(2) 向量组中任意$r+1$个向量(如果存在的话)都是线性相关的.

任意$r+1$个向量(如果存在的话)都可由极大无关组(r个向量)线性表示，肯定是线性相关的. 否则会有$r+1\leq r$，矛盾（P95 推论1）

向量组的秩为r，则任取向量组中r个线性无关的向量都构成极大无关组

命题13 向量组$\alpha_1,\alpha_2,\cdots,\alpha_t$线性相关的充分必要条件是$r\{\alpha_1,\alpha_2,\cdots,\alpha_t\}<t$.

向量组 $\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\subset \mathrm{F}^n$ 可由 $\{\beta_1,\beta_2,\cdots,\beta_t\}$ 线性表示 $\Longleftrightarrow$ $(\alpha_1,\alpha_2,\cdots,\alpha_s)=(\beta_1,\beta_2,\cdots,\beta_t)C_{t\times s}$

$A \quad = \quad BC$

$\{\alpha_1,\alpha_2,\cdots,\alpha_s\}$ 线性无关 $\Longleftrightarrow$ $(\alpha_1,\alpha_2,\cdots,\alpha_s)\begin{pmatrix}x_1\\x_2\\\vdots\\x_s\end{pmatrix}=AX=0$ 只有零解

$\Updownarrow$

另一种证明方法 $\quad r(A)\le r(C)\le \min\{s,t\} \qquad r(A)=s$

$r(C)=s, \quad s\le t$

$\{\alpha_1,\alpha_2,\cdots,\alpha_s\}$的极大无关组可由$\{\beta_1,\beta_2,\cdots,\beta_t\}$极大无关组线性表示

$\mathrm{r}\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\le \mathrm{r}(\beta_1,\beta_2,\cdots,\beta_t)$

$L\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\subseteq L(\beta_1,\beta_2,\cdots,\beta_t)$

如果 $\{\beta_1,\beta_2,\cdots,\beta_t\}$ 也可由 $\{\alpha_1,\alpha_2,\cdots,\alpha_s\}$ 线性表示 $\longrightarrow$

$L\{\alpha_1,\alpha_2,\cdots,\alpha_s\}=L(\beta_1,\beta_2,\cdots,\beta_t)$

$\mathrm{r}\{\alpha_1,\alpha_2,\cdots,\alpha_s\}=\mathrm{r}(\beta_1,\beta_2,\cdots,\beta_t)$

定理4 如果向量组$\alpha_1,\alpha_2,\cdots,\alpha_s$可由向量组$\beta_1,\beta_2,\cdots,\beta_t$线性表示，那么 $r\{\alpha_1,\alpha_2,\cdots,\alpha_s\}\le r\{\beta_1,\beta_2,\cdots,\beta_t\}$.

证明 设$r\{\alpha_1,\alpha_2,\cdots,\alpha_s\}=r$， $\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$是$\alpha_1,\alpha_2,\cdots,\alpha_s$的极大无关组；$r\{\beta_1,\beta_2,\cdots,\beta_t\}=u$， $\beta_{j_1},\beta_{j_2},\cdots,\beta_{j_u}$是$\beta_1,\beta_2,\cdots,\beta_t$的极大无关组．根据定理3的推论2，$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$可由$\beta_{j_1},\beta_{j_2},\cdots,\beta_{j_u}$线性表示. 因为$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_r}$线性无关，故由定理3的推论1得$r\le u$.

推论1　等价的向量组的秩相等.

推论2　如果$\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_s}$是向量组$\alpha_1,\alpha_2,\cdots,\alpha_t$中的部分向量构成的向量组，那么

$$r\{\alpha_{i_1},\alpha_{i_2},\cdots,\alpha_{i_s}\}\le r\{\alpha_1,\alpha_2,\cdots,\alpha_t\}.$$

例：设 $\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5$ 是 F^n 的向量. 已知

$\mathrm{r}(\alpha_1,\alpha_2,\alpha_3)=\mathrm{r}(\alpha_1,\alpha_2,\alpha_3,\alpha_4)=3,\quad \mathrm{r}(\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5)=4$

证明： $\alpha_1,\alpha_2,\alpha_3,2\alpha_4+\alpha_5$ 线性无关.

证： $\mathrm{r}(\alpha_1,\alpha_2,\alpha_3)=3,\ \rightarrow \alpha_1,\alpha_2,\alpha_3$ 线性无关.

$\mathrm{r}(\alpha_1,\alpha_2,\alpha_3,\alpha_4)=3,\ \rightarrow \alpha_1,\alpha_2,\alpha_3,\alpha_4$ 线性相关，α_4 可以由 $\alpha_1,\alpha_2,\alpha_3$ 线性表示.

$\mathrm{r}(\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5)=4,\quad \alpha_5$ 不可以由 $\alpha_1,\alpha_2,\alpha_3$ 线性表示,

(否则, $\mathrm{r}(\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5)=3$, 矛盾).

设 $k_1\alpha_1+k_2\alpha_2+k_3\alpha_3+k_4(2\alpha_4+\alpha_5)=0$

则 k_4 必为0，否则，

$$-\left(\frac{k_1}{k_4}\alpha_1+\frac{k_2}{k_4}\alpha_2+\frac{k_3}{k_4}\alpha_3+2\alpha_4\right)=\alpha_5$$

因为 α_4 可以由 $\alpha_1,\alpha_2,\alpha_3$ 线性表示, 所以 α_5 也可以由 $\alpha_1,\alpha_2,\alpha_3$ 线性表示, 矛盾. 所以, $k_4=0.$

$\alpha_1,\alpha_2,\alpha_3$ 线性无关, 所以, $k_1=k_2=k_3=0.$

所以, $\alpha_1,\alpha_2,\alpha_3,2\alpha_4+\alpha_5$ 线性无关.